

Priyanshu Pratik

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EDUCATION

Gati Shakti Vishwavidyalaya Bachelor of Technology in Artificial Intelligence and Data Science <i>Specialisation: Transportation and Logistics</i> - Current Status: 2nd Year (4th Semester) CGPA: 8.77 - Relevant Coursework: Machine Learning, Pattern Recognition, Design and Analysis of Algorithms, Signals and Systems, Computer Organisation and Architecture.	Vadodara, India Exp. 2028
Delhi Public School Class XII (Higher Secondary) - Percentage: 87.0% (435/500) - Joint Entrance Examination (JEE) Main: 96.6 Percentile (AIR: ~50,000)	Vadodara, India Year 2023
Soar Valley College Class X (Secondary / GCSE)	Leicester, United Kingdom Year 2020

TECHNICAL SKILLS

Languages: Python, Kotlin, Java, Javascript, C, MIPS Assembly, SQL
AI & Data Science: Machine Learning, Computer Vision, NLP, Pandas, PyTorch, TensorFlow Lite
Tools & Frameworks: Jetpack Compose, Streamlit, PyQt6, Git/GitHub, Docker
Core Concepts: Data Structures & Algorithms, DSP (FFT), MVVM Architecture, Object-Oriented Programming

PROJECTS

PulmoSense (Audio Breathing Analysis) <i>Kotlin, PyTorch, TFLite, Audio Processing</i>	Apr 2026
- Independently trained a custom PyTorch model and deployed it via TensorFlow Lite within an Android application to analyze respiratory audio patterns and deliver real-time health insights. - Engineered a custom processing engine (PulmoEngine) to compute 128-band Mel spectrograms from raw recordings, optimizing for real-time edge inference.	
Aura (Real-Time Audio Identifier) <i>Kotlin, Jetpack Compose, DSP</i>	Mar 2026 – Present
- Engineered a fault-tolerant Android audio identifier capable of real-time acoustic fingerprinting and "Query by Humming" melody matching using Fast Fourier Transform (FFT) algorithms. - Designed a reactive Jetpack Compose UI following MVVM architecture, integrating intelligent metadata parsing and platform-specific deep-linking for seamless song accessibility.	
Waveglider & Vector Squadron (3D Flight Simulations) <i>WebGL, Javascript, 3D Rendering</i>	Feb 2026 – Present
- Designed interactive, web-based 3D simulation games featuring user-controlled entities navigating vast, procedurally generated flight environments. - Optimized real-time rendering, dynamic third-person camera tracking, and physics mechanics to deliver smooth, low-latency browser-based gameplay.	
LocalPDF Pro (Desktop & Mobile Utility) <i>Python, Kotlin, PyQt6</i>	Dec 2025 – Present
- Deployed an open-source, 100% offline desktop application and corresponding Kotlin Android utility, providing secure, native PDF processing without cloud dependency. - Integrated an interactive analytics dashboard to track processed data volumes, prioritizing user privacy and standalone execution.	
Wayfinder (Pathfinding Visualiser) <i>Javascript, Algorithmic Routing</i>	Dec 2025 – Present
- Designed an interactive visualization platform for complex routing algorithms (A*, Dijkstra's), specifically optimized to model spatial traversal problems. - Demonstrated efficient map-based routing solutions directly applicable to large-scale transportation and logistics mapping systems.	
Beyond the Apex (F1 Telemetry Engine) <i>Python, Javascript, FastF1 API, Data Visualisation</i>	Nov 2025 – Present
Developed a telemetry analysis platform utilizing the FastF1 API, enabling users to generate and share 2D comparative visualizations for multi-driver session analysis under a tiered subscription model.	

PROBLEM SOLVING & ACADEMIC ENDEAVORS

Competitive Programming & Cryptic Hunts: Solved 160+ algorithmic problems on LeetCode and actively participate in complex puzzle-solving events (e.g., Cipher Saga) requiring rigorous analytical thinking. GATE 2027 Aspirant: Actively preparing for the GATE Data Science and Artificial Intelligence (DA) paper, demonstrating a commitment to rigorous academic standards in core AI domains.	
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INTERESTS

Space Exploration, Aeronautics, Astrophysics, Telemetry Analysis, Game Development.